

Electromagnetism & Electrodynamics HW9

Daniel Haim Breger, 316136944

Technion

(Date: July 17, 2026)

1 Spinning Sphere

1.1 Surface Current density

velocity on a rotating sphere is

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (1)$$

but in spherical

$$\boldsymbol{\omega} \times \mathbf{r} = \omega R \sin \theta \hat{\phi} \quad (2)$$

so

$$\mathbf{K} = \sigma \omega R \sin \theta \hat{\phi} \quad (3)$$

and the volumetric density is

$$\mathbf{j} = \sigma \omega R \sin \theta \delta(r - R) \hat{\phi} \quad (4)$$

1.2 magnetic dipole

we know that

$$\mathbf{m} = \frac{1}{2c} \int d^3x \mathbf{r} \times \mathbf{j} \quad (5)$$

since the volumetric integral will become a surface integral using the delta, we can say

$$\mathbf{m} = \frac{1}{2c} \oint \mathbf{r} \times \mathbf{K} dA \quad (6)$$

but

$$\mathbf{r} \times \mathbf{K} = \sigma \omega R^2 \sin \theta (\hat{\mathbf{r}} \times \hat{\phi}) = -\sigma \omega R^2 \sin \theta \hat{\theta} \quad (7)$$

but we know the moment must point in the $\hat{\mathbf{z}}$ direction here, so

$$(\mathbf{r} \times \mathbf{K}) \cdot \hat{\mathbf{z}} = \sigma \omega R^2 \sin^2 \theta \quad (8)$$

thus

$$m = \frac{\sigma \omega R^2}{2c} \oint \sin^2 \theta dA \quad (9)$$

$$= \frac{\sigma \omega \pi R^4}{c} \int_0^\pi \sin^3 \theta d\theta \quad (10)$$

$$= \frac{\sigma \omega \pi R^4}{c} \cdot \frac{4}{3} \quad (11)$$

$$= \frac{4\sigma \omega \pi R^4}{3c} \quad (12)$$

so

$$\mathbf{m} = \frac{4\pi \sigma \omega R^4}{3c} \hat{\mathbf{z}} \quad (13)$$

1.3 fields inside and outside

in TA 10 we saw that

$$\mathbf{A}(\mathbf{r}) = \frac{4\pi \sigma \omega R^3}{3c} \frac{r_{\leq}}{r_{\geq}^2} \sin \theta \hat{\phi} \quad (14)$$

which we can rewrite using the magnetic dipole

$$\mathbf{A}(\mathbf{r}) = \begin{cases} \frac{4\pi\sigma\omega R}{3c} r \sin\theta \hat{\phi} & r < R \\ \frac{4\pi\sigma\omega R^4}{3cr^2} \sin\theta \hat{\phi} & r > R \end{cases} \quad (15)$$

or

$$\mathbf{A}(\mathbf{r}) = \begin{cases} \frac{mr}{R^3} \sin\theta \hat{\phi} & r < R \\ \frac{m}{r^2} \sin\theta \hat{\phi} & r > R \end{cases} \quad (16)$$

The magnetic field is the rotor of the vector potential, so we know that, using the spherical rotor

$$B_r = \frac{1}{r \sin\theta} \partial_\theta (\sin\theta A_\phi), \quad B_\theta = -\frac{1}{r} \partial_r (r A_\phi), \quad B_\phi = 0 \quad (17)$$

Looking inside the sphere,

$$B_r = \frac{1}{r \sin\theta} \partial_\theta \left(m \frac{r}{R^3} \sin^2\theta \right) = \frac{2m}{R^3} \cos\theta \quad (18)$$

and

$$B_\theta = -\frac{1}{r} \partial_r \left(\frac{mr^2}{R^3} \sin\theta \hat{\phi} \right) = -\frac{2m}{R^3} \sin\theta \quad (19)$$

so

$$\mathbf{B} = \frac{2m}{R^3} (\cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\theta}) \quad (20)$$

but using the identity

$$\cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\theta} = \hat{\mathbf{z}} \quad (21)$$

$$\mathbf{B} = \frac{8\pi\sigma\omega R}{3c} \hat{\mathbf{z}} \quad (22)$$

and outside the sphere

$$B_r = \frac{1}{r \sin\theta} \partial_\theta \left(\frac{m}{r^2} \sin^2\theta \right) = 2 \frac{m}{r^3} \cos\theta \quad (23)$$

and

$$B_\theta = -\frac{1}{r} \partial_r \left(\frac{m}{r} \sin\theta \right) = \frac{m}{r^3} \sin\theta \quad (24)$$

so

$$\mathbf{B} = \frac{m}{r^3} (2 \cos\theta \hat{\mathbf{r}} + \sin\theta \hat{\theta}) \quad (25)$$

2 Magnetic Energy

2.1 two ways to write

we can rewrite

$$\iiint dV B^2 = \iiint dV \mathbf{B} \cdot \mathbf{B} = \iiint dV (\nabla \times \mathbf{A}) \cdot (\nabla \times \mathbf{A}) \quad (26)$$

and using the identity

$$\iiint dV B^2 = \iiint dV \nabla \cdot (\mathbf{A} \times (\nabla \times \mathbf{A})) + \iiint dV \mathbf{A} \cdot (\nabla \times (\nabla \times \mathbf{A})) \quad (27)$$

focusing on the first integral, we can use gauss to turn the volumetric integral on all space into a surface integral on a surface at infinity

$$\iiint dV \nabla \cdot (\mathbf{A} \times (\nabla \times \mathbf{A})) = \oint_{\infty} (\mathbf{A} \times (\nabla \times \mathbf{A})) = \oint_{\infty} (\mathbf{A} \times \mathbf{B}) \quad (28)$$

and since the currents are concentrated within a finite volume, at infinity this integral tends to zero, so we can ignore it. This leaves

$$\iiint dV B^2 = \iiint dV \mathbf{A} \cdot (\nabla \times \mathbf{B}) \quad (29)$$

but we can use Ampere's law which says

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{j} \quad (30)$$

so

$$\frac{1}{8\pi} \iiint dV B^2 = \frac{1}{2c} \iiint dV \mathbf{A} \cdot \mathbf{j} \quad (31)$$

as required. Next, using the coulomb gauge potential

$$\mathbf{A} = \frac{1}{c} \iiint \frac{\mathbf{j}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} dV' \quad (32)$$

inserting this

$$\frac{1}{2c} \iiint dV \mathbf{A} \cdot \mathbf{j} = \frac{1}{2c} \iiint dV \frac{1}{c} \iiint \frac{\mathbf{j}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} dV' \cdot \mathbf{j} \quad (33)$$

$$= \frac{1}{2c^2} \iiint dV \iiint \frac{\mathbf{j}(\mathbf{x}') \cdot \mathbf{j}(\mathbf{x})}{|\mathbf{x} - \mathbf{x}'|} dV' \quad (34)$$

as required.

2.2 two currents

The energy is

$$U = \frac{1}{2c} \iiint dV \mathbf{A} \cdot \mathbf{j} \quad (35)$$

inserting the split currents and fields

$$U = \frac{1}{2c} \iiint dV (\mathbf{A}_1 + \mathbf{A}_2) \cdot (\mathbf{j}_1 + \mathbf{j}_2) \quad (36)$$

$$= \frac{1}{2c} \iiint dV (\mathbf{j}_1 \cdot \mathbf{A}_1 + \mathbf{j}_1 \cdot \mathbf{A}_2 + \mathbf{j}_2 \cdot \mathbf{A}_1 + \mathbf{j}_2 \cdot \mathbf{A}_2) \quad (37)$$

But the interaction energy from TA9 is

$$\varepsilon = \frac{1}{2c} \iiint dx^3 \rho(\mathbf{x}) \phi_{\text{ext}}(\mathbf{x}) \quad (38)$$

in other words, we only care about the energy coming from the interaction of currents with external fields to itself, so elements of the form $\mathbf{j}_i \cdot \mathbf{A}_i$ can be ignored

$$\varepsilon = \frac{1}{2c} \iiint dV (\mathbf{j}_1 \cdot \mathbf{A}_2 + \mathbf{j}_2 \cdot \mathbf{A}_1) \quad (39)$$

but

$$\iiint dV \mathbf{j}_1 \cdot \mathbf{A}_2 = \frac{1}{c} \iiint dV \iiint \frac{\mathbf{j}_1(\mathbf{x}') \cdot \mathbf{j}_2(\mathbf{x})}{|\mathbf{x} - \mathbf{x}'|} dV' \quad (40)$$

but we can see that replacing $\mathbf{x} \leftrightarrow \mathbf{x}'$ gives us the same expression with the currents replaced, so

$$\iiint dV \mathbf{j}_1 \cdot \mathbf{A}_2 = \iiint dV \mathbf{j}_2 \cdot \mathbf{A}_1 \quad (41)$$

so

$$\varepsilon = \frac{1}{c} \iiint dV \mathbf{j}_1 \cdot \mathbf{A}_2 = \frac{1}{c} \iiint dV \mathbf{j}_2 \cdot \mathbf{A}_1 \quad (42)$$

as required.

2.3 magnetic dipole interaction energy

The potential from a dipole is

$$\mathbf{A} = \frac{\mathbf{m} \times \mathbf{r}}{r^3} \quad (43)$$

so

$$\varepsilon = \frac{1}{c} \iiint dV \mathbf{j}_{\text{ext}} \cdot \mathbf{A} = \frac{1}{c} \iiint dV \frac{1}{r^3} \mathbf{j}_{\text{ext}} \cdot (\mathbf{m} \times \mathbf{r}) \quad (44)$$

using the identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$

$$\varepsilon = \frac{1}{c} \iiint dV \frac{1}{r^3} \mathbf{m} \cdot (\mathbf{r} \times \mathbf{j}_{\text{ext}}) = \mathbf{m} \cdot \left(\frac{1}{c} \iiint dV \frac{\mathbf{r} \times \mathbf{j}_{\text{ext}}}{r^3} \right) \quad (45)$$

recalling from ancient history (physics 2p) the bio-savart law

$$\mathbf{B}(0) = \frac{1}{c} \iiint dV \frac{\mathbf{j}_{\text{ext}} \times (-\mathbf{x})}{r^3} \quad (46)$$

and using another vector identity $\mathbf{A} \times (-\mathbf{B}) = \mathbf{B} \times \mathbf{A}$

$$\mathbf{B}_{\text{ext}}(0) = \frac{1}{c} \iiint dV \frac{\mathbf{x} \times \mathbf{j}_{\text{ext}}}{r^3} \quad (47)$$

so

$$U = \mathbf{m} \cdot \mathbf{B}_{ext}(0) \quad (48)$$

2.4 Radial component

Since there are no magnetic monopoles (don't tell the high energy people)

$$\nabla \cdot \mathbf{B} = 0 \quad (49)$$

or in cylindrical

$$\frac{1}{r} \partial_r (r B_r) + \partial_z B_z = 0 \quad (50)$$

but

$$B_z = B_0 z / L \quad (51)$$

so

$$\frac{1}{r} \partial_r (r B_r) + B_0 / L = 0 \quad (52)$$

so

$$B_r = -\frac{B_0}{2L} r + C/r \quad (53)$$

demanding no infinities $C = 0$ so

$$B_r = -\frac{B_0}{2L} r \quad (54)$$

2.5 potential energy

we're given

$$dU = -\mathbf{m} \cdot d\mathbf{B} = -\alpha \mathbf{B} \cdot d\mathbf{B} \quad (55)$$

so

$$U = -\frac{\alpha}{2} \mathbf{B}^2 + C \quad (56)$$

since this is a potential the choice of C is arbitrary so $C = 0$ because I say so

$$U = -\frac{\alpha}{2} \mathbf{B}^2 \quad (57)$$

but we know that

$$\mathbf{B} = \frac{B_0 z}{L} \hat{\mathbf{z}} - \frac{B_0 r}{2L} \hat{\mathbf{r}} \quad (58)$$

so

$$B^2 = \frac{B_0^2 z^2}{L^2} + \frac{B_0^2 r^2}{4L^2} = \frac{B_0^2}{L^2} \left(z^2 + \frac{r^2}{4} \right) \quad (59)$$

so

$$U = -\frac{\alpha B_0^2}{2L^2} \left(z^2 + \frac{r^2}{4} \right) \quad (60)$$

2.6 equilibrium

The equilibrium will be where the force on the particle is zero, so we need

$$-\nabla U = 0 \quad (61)$$

but we notice immediately that after taking the gradient the only equilibrium point would be

$$r = 0, \quad z = 0 \quad (62)$$

for it to be stable, we want the potential to rise around the point so

$$\alpha < 0 \quad (63)$$

and recalling the potential energy of a harmonic oscillator

$$U = \frac{1}{2}M\omega^2x^2 \quad (64)$$

we can equate it and find

$$\frac{1}{2}M\omega_z^2z^2 = -\frac{\alpha B_0^2}{2L^2}z^2 \quad (65)$$

therefore

$$\omega_z^2 = -\frac{\alpha B_0^2}{ML^2} \quad (66)$$

and

$$\frac{1}{2}M\omega_r^2r^2 = -\frac{\alpha B_0^2}{2L^2}\frac{r^2}{4} \quad (67)$$

so

$$\omega_r^2 = -\frac{\alpha B_0^2}{4ML^2} \quad (68)$$